

The cross-rank obstruction for ribbon operators does not need a single parameter: free height weights force Murnaghan–Nakayama

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Abstract

For $e \geq 1$ let $R_e(t)$ be the operator on the ring of symmetric functions that adds a connected e -ribbon with weight t^{ht} . I proved previously that for $e \neq f$ the operators $R_e(t)$ and $R_f(t)$ commute only at $t = -1$. That theorem was proved for a *single* parameter t , and the height weight t^N was assumed multiplicative in N . Following LLT’s 1995 move — replace a parameter raised to a statistic by a parameter indexed by that statistic — I remove the assumption entirely: give each height value N its own weight w_N , with no relation between them, and allow the weight functions of R_e and R_f to be *unrelated to each other* as well.

Result. The obstruction survives, and it survives in the strongest form the question admits. For $1 \leq e < f$ and arbitrary weight functions $w : \{0, \dots, e-1\} \rightarrow K$, $\bar{w} : \{0, \dots, f-1\} \rightarrow K$ over an integral domain, $[R_e^w, R_f^{\bar{w}}] = 0$ if and only if one of the operators is zero or

$$w_N = (-1)^N w_0 \quad (N \leq e-1), \quad \bar{w}_N = (-1)^N \bar{w}_0 \quad (N \leq f-1),$$

i.e. both are scalar multiples of multiplication by a power sum. *Every parameter the pair can see is pinned by the pair.* In the coordinates of the brief ($\tau_N = (-1)^N w_N$, so that single- t is $\tau_N = (-t)^N$) the commuting locus is the single point $\tau \equiv 1$. This is outcome 1: the single- t hypothesis of the forcing theorem was never load-bearing, and the theorem strengthens to arbitrary height weights.

The proof isolates *why*. The commutator splits into a one-bead and a two-bead sector, and the two sectors cut out different loci: the two-bead sector alone forces both weights to be geometric with reciprocal ratios ($w_N = s^N$, $\bar{w}_N = s^{-N}$), which is exactly the restoration of a single parameter; the one-bead sector alone forces the anchor outright. The mechanism in the one-bead sector is a dichotomy internal to the height statistic: an interior site of a long ribbon is counted by the height on the route that steps over it and not counted on the route that factors the ribbon there. The two readings differ by exactly one, so closure forces $w_{N+1} = -w_N$, and the familiar scalar $1 + t$ is only the instance $N = 0$.

1 Conventions, and the two things that had to be pinned first

Convention 1. K is an integral domain; Λ_K is the ring of symmetric functions over K with Schur basis $\{s_\lambda\}$ and Hall inner product $\langle s_\lambda, s_\mu \rangle = \delta_{\lambda\mu}$. For a partition λ , $M(\lambda) = \{\lambda_j - j : j \geq 1\}$ is its Maya set (beads); $m(x) = [x \in M]$. For $x < y$ we write $\#(M \cap (x, y))$ for the number of beads strictly between x and y . For a connected border strip (ribbon) μ/λ , $\text{ht}(\mu/\lambda)$ is its number of rows minus one.

Convention 2 (the deformed operator). Fix $e \geq 1$ and a *weight function* $w : \{0, 1, \dots, e-1\} \rightarrow K$, written w_N . Define $R_e^w : \Lambda_K \rightarrow \Lambda_K$ by

$$R_e^w s_\lambda = \sum_{\mu} w_{\text{ht}(\mu/\lambda)} s_\mu,$$

the sum over $\mu \supset \lambda$ with μ/λ a connected e -ribbon. (The height of an e -ribbon lies in $\{0, \dots, e-1\}$, so R_e^w sees exactly these e values and no more.) The classical operator is $w_N = t^N$: $R_e^{(t^\bullet)} = R_e(t)$.

Two conventions had to be settled before any deformation, because getting either wrong silently answers a different question.

First: the sign is not a second parameter

The fermionic normal form of [1] reads $R_e(t) = \sum_b \psi_{b+e} \psi_b^* (-t)^{N_{(b,b+e)}}$ with $N_{(b,b+e)} = \sum_{b < j < b+e} n_j$, and the deformation proposed in the brief is $R_e^\tau = \sum_b \psi_{b+e} \psi_b^* \tau_{N_{(b,b+e)}}$. The $(-1)^N$ inside $(-t)^N$ is *not* part of the weight: it cancels the Jordan–Wigner sign. Explicitly [1, Lem. 5], $\psi_{b+e} \psi_b^* |M\rangle = (-1)^{\#(M \cap (b,b+e))} |M'\rangle$, so

$$R_e^\tau |\lambda\rangle = \sum_\mu (-1)^{\text{ht}(\mu/\lambda)} \tau_{\text{ht}(\mu/\lambda)} |\mu\rangle, \quad \text{that is} \quad \boxed{w_N = (-1)^N \tau_N}.$$

The map $\tau \mapsto w$ is a diagonal linear isomorphism of the parameter space, so a vanishing locus computed in one coordinate transports verbatim to the other; there is no double counting and nothing extra to deform. Under it:

$$\text{single-}t : \tau_N = (-t)^N \iff w_N = t^N; \quad \text{anchor: } \tau \equiv 1 \iff w_N = (-1)^N \iff t = -1.$$

I work throughout on the partition side, in the coordinates w , where no fermion sign occurs at all, and translate to τ only in Corollary 9. This makes the hazard structurally impossible rather than merely avoided.

Second: shared versus independent weights

LLT's move gives one parameter per height value, *shared* across all ribbon sizes: $w^{(e)} = w^{(f)}$. The outer bound gives each R_e its own weight function. I prove the theorem for **independent** weight functions; the shared statement is the special case. Nothing below assumes any relation between w and $\bar{w}$.

What is being tested

Question 3. For which pairs $(w, \bar{w})$ does $[R_e^w, R_f^{\bar{w}}] = 0$, $e \neq f$?

The point of the question is that R_e^w has $e-1$ free parameters after normalisation, so $e \leq 2$ cannot distinguish a free weight from a single t : R_1^w has none and R_2^w has one, exactly as many as t . The first genuinely new freedom is w_2 versus w_1^2 at $e=3$; accordingly $[R_1, R_2]$ is a degenerate test and is not used anywhere below.

2 Realisability: every local bead pattern occurs

Everything is decided inside one window of length $e+f$. The following says the window may be prescribed arbitrarily.

Lemma 4. *Let $g \geq 2$ and let $S \subseteq \{1, \dots, g-1\}$. Then there is a partition λ with $0 \in M(\lambda)$, $g \notin M(\lambda)$ and $M(\lambda) \cap (0, g) = S$.*

Proof. Put $r = 1 + |S|$ and $M = \mathbb{Z}_{\leq -r-1} \cup \{0\} \cup S$. Then M contains all sufficiently negative integers and is bounded above, and its charge is $\#(M \cap \mathbb{Z}_{\geq 0}) - \#(\mathbb{Z}_{<0} \setminus M) = (1 + |S|) - r = 0$; hence $M = M(\lambda)$ for a unique partition λ . By construction $0 \in M$, $g \notin M$ (as $S \subseteq \{1, \dots, g-1\}$) and $M \cap (0, g) = S$. The r deleted sites lie below 0 and so do not meet the window. $\square$

3 The one-bead matrix element with free weights

Fix $1 \leq e < f$ and put $g = e + f$. Throughout, $M = M(\lambda)$ is a Maya set with $b \in M$ and $b + g \notin M$, and μ is the partition with $M(\mu) = (M \setminus \{b\}) \cup \{b + g\}$. Write

$$m_e = m(b + e), \quad m_f = m(b + f),$$

$$P_e = \#(M \cap (b, b + e)), \quad S_e = \#(M \cap (b + e, b + g)), \quad P_f = \#(M \cap (b, b + f)), \quad S_f = \#(M \cap (b + f, b + g)).$$

Lemma 5 (one-bead element). *With the notation above,*

$$\langle s_\mu, [R_e^w, R_f^{\bar{w}}] s_\lambda \rangle = (1 - 2m_f) w_{S_f} \bar{w}_{P_f} - (1 - 2m_e) w_{P_e} \bar{w}_{S_e}.$$

All four subscripts are legal: $P_e, S_f \leq e - 1$ and $P_f, S_e \leq f - 1$.

Proof. By Convention 2 and [1, Lem. 4], $R_g^\bullet$ acts on Maya sets by legal bead moves ($b' \in M$, $b' + g \notin M$), with the height of the resulting ribbon equal to the number of beads strictly jumped. Hence

$$\langle s_\mu, R_f^{\bar{w}} R_e^w s_\lambda \rangle = \sum_{\text{routes}} w_{h_1} \bar{w}_{h_2},$$

over pairs of legal moves $M \xrightarrow{(x,e)} M_1 \xrightarrow{(y,f)} M(\mu)$, with $h_1 = \#(M \cap (x, x + e))$ and $h_2 = \#(M_1 \cap (y, y + f))$.

Which routes exist. We have $M(\mu) = ((M \setminus \{x\}) \cup \{x + e\}) \setminus \{y\} \cup \{y + f\}$. If no removed site coincides with an added one then $|M \triangle M(\mu)| = 4$; but $|M \triangle M(\mu)| = 2$ here. Since $x \neq x + e$ and $y \neq y + f$, the only possibilities are $y = x + e$ or $x = y + f$.

Route B ($y = x + e$). Then $M(\mu) = (M \setminus \{x\}) \cup \{x + g\}$, so $x = b$, $y = b + e$. Legality of the first move needs $b \in M$ (given) and $b + e \notin M$, i.e. $m_e = 0$; the second move is then automatically legal ($b + e \in M_1$, and $b + g \notin M_1$ because $b + g \notin M$ and M_1 added only $b + e$). Its heights: $h_1 = \#(M \cap (b, b + e)) = P_e$, and since M_1 differs from M only at b and $b + e$, both outside the *open* interval $(b + e, b + g)$, $h_2 = \#(M \cap (b + e, b + g)) = S_e$. Contribution $w_{P_e} \bar{w}_{S_e}$, present iff $m_e = 0$.

Route C ($x = y + f$). Then $M(\mu) = (M \setminus \{y\}) \cup \{y + g\}$, so $y = b$, $x = b + f$. Legality of the first move needs $b + f \in M$, i.e. $m_f = 1$, and $b + g \notin M$ (given); the second is then legal. Heights: $h_1 = \#(M \cap (b + f, b + g)) = S_f$, and M_1 differs from M only at $b + f$ and $b + g$, both outside $(b, b + f)$, so $h_2 = \#(M \cap (b, b + f)) = P_f$. Contribution $w_{S_f} \bar{w}_{P_f}$, present iff $m_f = 1$.

Therefore $\langle s_\mu, R_f^{\bar{w}} R_e^w s_\lambda \rangle = (1 - m_e) w_{P_e} \bar{w}_{S_e} + m_f w_{S_f} \bar{w}_{P_f}$. Running the same enumeration for the other order — now the f -move is performed first — Route B' (f -move at b , then e -move at $b + f$) exists iff $m_f = 0$ and contributes $\bar{w}_{P_f} w_{S_f}$; Route C' (f -move at $b + e$, then e -move at b) exists iff $m_e = 1$ and contributes $\bar{w}_{S_e} w_{P_e}$. Hence $\langle s_\mu, R_e^w R_f^{\bar{w}} s_\lambda \rangle = (1 - m_f) w_{S_f} \bar{w}_{P_f} + m_e w_{P_e} \bar{w}_{S_e}$, and subtracting gives the boxed formula. The subscript bounds hold because P_e, S_f count beads in open intervals of length e and P_f, S_e in open intervals of length f . $\square$

Remark 6 (what the formula says, in words). The two monomials are the two ways of *factoring* the long ribbon $b \rightarrow b + g$ at an interior site: through the hinge $b + e$, or through the hinge $b + f$. Remarkably, the pair of heights (P_e, S_e) of the factorisation at $b + e$ is the same whether that site is occupied or empty — what the occupancy changes is *which order realises it*: if $b + e$ is empty the e -move goes first, and if $b + e$ is occupied its resident bead must vacate first, so the f -move goes first. The monomial therefore appears on one side of the commutator or the other according to m_e , which is the factor $(1 - 2m_e)$. This is the whole mechanism, and the next section is bookkeeping.

Specialising $w = \bar{w} = (t^\bullet)$ recovers $(m_e - m_f)(1 + t)t^{N-1}$ with $N = \#(M \cap (b, b + g))$, since $P_e + S_e = N - m_e$ and $P_f + S_f = N - m_f$: this is [2, Thm. 5(1)].

4 The forcing theorem

Theorem 7 (free height weights force Murnaghan–Nakayama). *Let $1 \leq e < f$, let K be an integral domain, and let $w : \{0, \dots, e-1\} \rightarrow K$ and $\bar{w} : \{0, \dots, f-1\} \rightarrow K$ be arbitrary. Then $[R_e^w, R_f^{\bar{w}}] = 0$ if and only if one of the following holds:*

- (a) $w \equiv 0$ (equivalently $R_e^w = 0$);
- (b) $\bar{w} \equiv 0$ (equivalently $R_f^{\bar{w}} = 0$);
- (c) $w_N = (-1)^N w_0$ for $0 \leq N \leq e-1$ and $\bar{w}_N = (-1)^N \bar{w}_0$ for $0 \leq N \leq f-1$.

In case (c), $R_e^w = w_0 M_{p_e}$ and $R_f^{\bar{w}} = \bar{w}_0 M_{p_f}$, multiplication by power sums.

Only one of the four occupancy patterns of Lemma 5 is used, namely $m_e = 1, m_f = 0$; and only the one-bead sector is used. Write it out. By Lemma 4 we may choose the window freely, so let

$$u = \#(M \cap (b, b+e)), \quad v = \#(M \cap (b+e, b+f)), \quad z = \#(M \cap (b+f, b+g)),$$

which range independently over $0 \leq u \leq e-1, 0 \leq v \leq f-e-1, 0 \leq z \leq e-1$ (the three open intervals have $e-1, f-e-1$ and $g-f-1 = e-1$ sites). With $m_e = 1, m_f = 0$ we get $P_e = u, S_e = v+z, P_f = u+1+v, S_f = z$, and Lemma 5 gives the element $w_z \bar{w}_{u+v+1} + w_u \bar{w}_{v+z}$. So:

Lemma 8 (the forcing relation). *If $[R_e^w, R_f^{\bar{w}}] = 0$ then*

$$w_z \bar{w}_{u+v+1} + w_u \bar{w}_{v+z} = 0 \quad \text{for all } 0 \leq u, z \leq e-1, \quad 0 \leq v \leq f-e-1. \quad (1)$$

Proof. Immediate from Lemmas 4 and 5: given (u, v, z) , choose $S \subseteq \{1, \dots, g-1\}$ with $e \in S, f \notin S, |S \cap (0, e)| = u, |S \cap (e, f)| = v, |S \cap (f, g)| = z$, and take λ from Lemma 4 and $b = 0$. $\square$

Proof of Theorem 7. Sufficiency. If $w \equiv 0$ or $\bar{w} \equiv 0$ one of the operators is zero and the commutator vanishes. Suppose (c). Both operators are K -linear in their weights, so we may assume $w_0 = \bar{w}_0 = 1$, i.e. $w_N = \bar{w}_N = (-1)^N$ throughout their ranges. Then in the one-bead sector, using $P_e + S_e = N - m_e$ and $P_f + S_f = N - m_f$ with $N = \#(M \cap (b, b+g))$, Lemma 5 gives

$$(1 - 2m_f)(-1)^{N-m_f} - (1 - 2m_e)(-1)^{N-m_e} = (-1)^N - (-1)^N = 0,$$

because $(1 - 2m)(-1)^{-m} = 1$ for $m \in \{0, 1\}$. In the two-bead sector (Lemma 12 below) the element is $\bar{w}_Q w_{P-k} - w_P \bar{w}_{Q+k} = (-1)^{P+Q-k} - (-1)^{P+Q+k} = 0$. By Lemma 11 these are all the matrix elements, so the commutator vanishes.

Necessity. Assume $[R_e^w, R_f^{\bar{w}}] = 0$, so (1) holds.

Case 1: $w_0 = 0$ and $w \not\equiv 0$. Let i be least with $w_i \neq 0$; then $1 \leq i \leq e-1$. Taking $z = 0$ in (1) gives $w_u \bar{w}_v = 0$ for all $u \leq e-1, v \leq f-e-1$; with $u = i$ and K a domain this forces

$$\bar{w}_v = 0 \quad (0 \leq v \leq f-e-1). \quad (2)$$

Now take $z = i$ in (1): $w_i \bar{w}_{u+v+1} = -w_u \bar{w}_{v+i}$. For $u < i$ the right side is 0, so $\bar{w}_n = 0$ for all n of the form $u+v+1$ with $u \leq i-1, v \leq f-e-1$, i.e. for $1 \leq n \leq i+f-e-1$; together with (2),

$$\bar{w}_n = 0 \quad (0 \leq n \leq i+f-e-1). \quad (3)$$

For $i \leq u \leq e-1$ the right side is $-w_u \bar{w}_{v+i}$ and $v+i \leq i+f-e-1$, so $\bar{w}_{v+i} = 0$ by (3); hence $\bar{w}_{u+v+1} = 0$ for $i \leq u \leq e-1, 0 \leq v \leq f-e-1$, i.e. for $i+1 \leq n \leq f-1$. With (3) this gives $\bar{w} \equiv 0$: case (b).

Case 2: $\bar{w}_0 = 0$ and $w_0 \neq 0$. Taking $z = 0, v = 0$ in (1) gives $w_0 \bar{w}_{u+1} = -w_u \bar{w}_0 = 0$, so $\bar{w}_n = 0$ for $1 \leq n \leq e$; with $\bar{w}_0 = 0, \bar{w}_n = 0$ for $n \leq e$. Inductively, suppose $\bar{w}_n = 0$ for all $n \leq m$ with $m \geq e$ and $m+1 \leq f-1$. Write $m = u+v$ with $0 \leq u \leq e-1, 0 \leq v \leq f-e-1$ (possible since $m \leq f-2 = (e-1) + (f-e-1)$). Then $v \leq m$, so (1) with $z = 0$ gives $w_0 \bar{w}_{m+1} = -w_u \bar{w}_v = 0$, whence $\bar{w}_{m+1} = 0$. So $\bar{w} \equiv 0$: case (b).

Case 3: $w_0 \neq 0$ and $\bar{w}_0 \neq 0$. Rescaling, assume $w_0 = \bar{w}_0 = 1$. Two specialisations of (1):

$$(B0) \quad \bar{w}_{u+v+1} = -w_u \bar{w}_v \quad (z = 0), \quad (B1) \quad w_z \bar{w}_{v+1} = -\bar{w}_{v+z} \quad (u = 0).$$

From (B0) at $u = v = 0$: $\bar{w}_1 = -1$. From (B1) at $v = 0$: $w_z \bar{w}_1 = -\bar{w}_z$, i.e.

$$w_z = \bar{w}_z \quad (0 \leq z \leq e-1). \quad (4)$$

From (B0) at $v = 0$ and (4): $\bar{w}_{u+1} = -w_u = -\bar{w}_u$ for $0 \leq u \leq e-1$, so $\bar{w}_n = (-1)^n$ for $n \leq e$. From (B0) at $u = 0$: $\bar{w}_{v+1} = -\bar{w}_v$ for $0 \leq v \leq f-e-1$, so $\bar{w}_n = (-1)^n$ for $n \leq f-e$. Let n satisfy $\max(e, f-e) < n \leq f-1$ and assume $\bar{w}_m = (-1)^m$ for all $m < n$. Write $n-1 = u+v$ with $0 \leq u \leq e-1, 0 \leq v \leq f-e-1$ (possible since $n-1 \leq f-2$). Both $u \leq e-1 < n$ and $v \leq f-e-1 < n$, so by (B0) and (4)

$$\bar{w}_n = -w_u \bar{w}_v = -\bar{w}_u \bar{w}_v = -(-1)^{u+v} = (-1)^{u+v+1} = (-1)^n.$$

Hence $\bar{w}_n = (-1)^n$ for all $0 \leq n \leq f-1$, and $w_n = \bar{w}_n = (-1)^n$ for $0 \leq n \leq e-1$ by (4): case (c) with $w_0 = \bar{w}_0 = 1$.

Finally, in case (c) with $w_0 = 1, R_e^w s_\lambda = \sum_\mu (-1)^{\text{ht}(\mu/\lambda)} s_\mu$, which is the Murnaghan–Nakayama rule for $p_e s_\lambda$ (Remark 16). $\square$

Corollary 9 (answer to the primary question, in τ coordinates). *Let $\tau_N = (-1)^N w_N$ be the fermionic weights of the brief, normalised by $\tau_0 = 1$, shared across e . For any $e \neq f$, the locus $\{\tau : [R_e^\tau, R_f^\tau] = 0\}$ is the single point $\tau_N = 1$ ($0 \leq N \leq \max(e, f) - 1$). In particular for $(e, f) = (1, 3)$ — the first pair that can distinguish multi- t from single- t , since R_3^τ is the first operator carrying two free parameters — the locus in the (τ_1, τ_2) -plane is the point $(1, 1)$. It meets the single- t parabola $\tau_2 = \tau_1^2$ only there, and it contains nothing else: there is no new commuting family off the single- t curve.*

Proof. Theorem 7 case (c) with $w_0 = \bar{w}_0 = 1$ reads $\tau_N = 1$; cases (a), (b) are excluded by $\tau_0 = 1$. For $(e, f) = (1, 3)$ relation (1) has $u = z = 0$ and $v \in \{0, 1\}$, giving $\bar{w}_1 = -1$ and $\bar{w}_2 = -\bar{w}_1 = 1$, i.e. $\tau_1 = \tau_2 = 1$. $\square$

Corollary 10 (the family). *Let $E \subseteq \mathbb{Z}_{\geq 1}$ with $|E| \geq 2$ and let $R_e^{w^{(e)}}$, $e \in E$, be nonzero deformed ribbon operators that pairwise commute. Then for every $e \in E$, $w_N^{(e)} = (-1)^N w_0^{(e)}$ and $R_e^{w^{(e)}} = w_0^{(e)} M_{p_e}$. In particular the only commutative algebra generated by deformed ribbon operators of at least two distinct sizes is the algebra of power-sum multiplications.*

5 Why: the two sectors cut out different loci

Theorem 7 used only the one-bead sector. The two-bead sector gives an independent second route to (most of) the conclusion, and comparing the two says exactly where the extra freedom of a free weight dies.

Lemma 11 (sectors). $\langle s_\mu, [R_e^w, R_f^{\bar{w}}] s_\lambda \rangle = 0$ unless $|M(\lambda) \triangle M(\mu)| \in \{2, 4\}$.

Proof. Each $R_g^\bullet$ moves one bead, so a product moves at most two; and $M(\mu) = M(\lambda)$ is impossible since $|\mu| = |\lambda| + e + f > |\lambda|$. $\square$

Lemma 12 (two-bead element). *Let $e \neq f$ and $M(\mu) = (M \setminus \{b, c\}) \cup \{b+e, c+f\}$ with $b, c \in M$, $b+e, c+f \notin M$ and $b, c, b+e, c+f$ pairwise distinct. Put $P = \#(M \cap (b, b+e))$, $Q = \#(M \cap (c, c+f))$ and $k = [b+e \in (c, c+f)] - [b \in (c, c+f)] \in \{-1, 0, 1\}$. Then*

$$\langle s_\mu, [R_e^w, R_f^{\bar{w}}] s_\lambda \rangle = \bar{w}_Q w_{P-k} - w_P \bar{w}_{Q+k}.$$

Every other $|M \triangle M(\mu)| = 4$ element vanishes.

Proof. This is [2, Thm. 5(2)] with the weights left free; the route enumeration is unchanged. No cancellation occurs, so the removed sites are $M \setminus M(\mu)$ and the added sites are $M(\mu) \setminus M$; for $e \neq f$ the pair (b, c) is determined (b is the unique removed site with $b+e$ added). Performing the e -move first gives heights P and $Q+k$ (the e -move changes the count in $(c, c+f)$ by k); performing the f -move first gives Q and $P-k$ (by [2, Lem. 4], the two interference indices sum to zero off the diagonals $b=c$, $b+e=c+f$). $\square$

Proposition 13 (the two-bead sector restores a single parameter, and only that). *Let $2 \leq e < f$ and $w_0 = \bar{w}_0 = 1$. If every two-bead matrix element of $[R_e^w, R_f^{\bar{w}}]$ vanishes, then there is $s \in K^\times$ with*

$$w_N = s^N \quad (N \leq e-1), \quad \bar{w}_N = s^{-N} \quad (N \leq f-1).$$

If in addition the weights are shared ($w = \bar{w}$ on the common range and $e \geq 2$), then $s^2 = 1$. Conversely every such pair kills the two-bead sector.

Proof. Take $b=0$ and $c=d$ with $1 \leq d \leq e-1$; then $b < c < b+e < c+f$ (the last because $d \geq 1 > e-f$), so the four sites are distinct, $b+e \in (c, c+f)$, $b \notin (c, c+f)$ and $k=1$. Decompose $(0, e) = (0, d) \sqcup \{d\} \sqcup (d, e)$ and $(d, d+f) = (d, e) \sqcup \{e\} \sqcup (e, d+f)$. Prescribing (Lemma 4) a beads in $(0, d)$, m beads in (d, e) , q beads in $(e, d+f)$ and none at e or $d+f$, we get $P = a+1+m$ and $Q = m+q$, with $a \leq d-1$, $m \leq e-d-1$, $q \leq d+f-e-1$. Choosing $d=e-1$ forces $m=0$ and leaves $a \leq e-2$, $q \leq f-2$ free and independent. The vanishing condition $\bar{w}_Q w_{P-1} = w_P \bar{w}_{Q+1}$ becomes

$$\bar{w}_q w_a = w_{a+1} \bar{w}_{q+1} \quad (0 \leq a \leq e-2, 0 \leq q \leq f-2).$$

At $a=q=0$: $1 = w_1 \bar{w}_1$, so $s := w_1$ is a unit with $\bar{w}_1 = s^{-1}$. At $a=0$: $\bar{w}_q = s \bar{w}_{q+1}$, so $\bar{w}_n = s^{-n}$ for $n \leq f-1$. At $q=0$: $w_a = w_{a+1} s^{-1}$, so $w_n = s^n$ for $n \leq e-1$. If the weights are shared then $s = w_1 = \bar{w}_1 = s^{-1}$. The converse is the computation $\bar{w}_Q w_{P-k} - w_P \bar{w}_{Q+k} = s^{-Q+P-k} - s^{P-Q-k} = 0$. $\square$

Remark 14 (the shape of the answer). Proposition 13 is the precise sense in which the multi- t freedom is an illusion, and it is worth separating from Theorem 7:

- The **two-bead sector** forces both weights to be *multiplicative*: $w_N = s^N$, $\bar{w}_N = s^{-N}$. That is precisely the statement $\tau_2 = \tau_1^2$ and its higher analogues — a single parameter, restored. It does *not* determine s .
- The **one-bead sector** forces the anchor outright, with no multiplicativity input at all (Theorem 7).

So the two mechanisms are genuinely different, and the one-bead mechanism is the stronger of the two. Note also that the two-bead sector is empty when $\min(e, f) = 1$ ([2, Cor. 9]: $k \neq 0$ forces one of the four sites to collide), which is why the theorem must rest on the one-bead sector: the pair $(1, 3)$ — the cheapest genuine test — has no two-bead sector at all.

Remark 15 (the mechanism, stated once more without indices). Read (1) at $u = z = 0$: $\bar{w}_{v+1} = -\bar{w}_v$. On the left, the site $b + e$ is a bead *jumped over* by the long f -move $b \rightarrow b + f$, and so is counted by its height. On the right, the same site is the *hinge* at which the $(e + f)$ -ribbon factors, and a hinge is an endpoint of both open intervals, hence counted by neither height. One occupied site, two readings, differing by exactly 1. Closure of the bracket demands that the weight convert that unit shift into a sign, and the only weights that can do so are the alternating ones. The scalar $1 + t$ of the single-parameter theorem is the instance $v = 0$ of this relation — it was never a fact about t , and giving t more room to move was never going to help.

6 Computational verification

All code is in `probes/2026-09-11-c2-Q147/`. The engine (`engine.py`) was written fresh for this note; in particular it never forms $(-t)^N$ or t^{ht} , and the weight enters only as an opaque symbol.

1. **Two independent ribbon enumerations agree.** `ribbons_young` enumerates all $\mu \vdash |\lambda| + e$ containing λ and tests the skew shape for the 2×2 and connectivity conditions, computing the height as $(\text{rows}) - 1$; `ribbons_maya` performs bead moves $b \rightarrow b + e$ and computes the height as beads jumped. They agree as multisets of (shape, height) for every λ with $|\lambda| \leq 8$ and every $e \leq 5$.
2. **Lemma 5 verified.** For $(e, f) \in \{(1, 2), (1, 3), (2, 3), (2, 4), (3, 4), (2, 5), (3, 5)\}$, all four occupancy patterns (m_e, m_f) and every bead pattern in the three open sub-intervals — 316 configurations, built by the construction of Lemma 4 — the formula agrees with the engine’s commutator on the nose, with *independent* symbolic weights. 0 mismatches.
3. **Lemma 12 verified** likewise on 42 configurations ($k = +1, (e, f) \in \{(2, 3), (2, 4), (3, 4), (3, 5)\}$). 0 mismatches.
4. **The locus, computed without the theorem.** Collecting every nonzero entry of $[R_e^w, R_f^{\bar{w}}]s_\lambda$ over all λ up to size 13 and solving the resulting polynomial system:

(e, f)	shared weights	independent weights
$(1, 2)$	$w_1 = -1$	—
$(1, 3)$	$w_1 = -1, w_2 = 1$	$\bar{w}_1 = -1, \bar{w}_2 = 1$
$(1, 4), (1, 5)$	$w_N = (-1)^N, N \leq f - 1$	—
$(2, 3), (2, 4), (2, 5), (2, 6)$	$w_N = (-1)^N, N \leq f - 1$	$w_N = \bar{w}_N = (-1)^N$
$(3, 4), (3, 5), (3, 6), (4, 5)$	$w_N = (-1)^N, N \leq f - 1$	$w_N = \bar{w}_N = (-1)^N$

In every case the solution set is the single point of Theorem 7(c), and every symbol visible to the pair appears in the equations.

5. **The sector split verified.** Separating the equations by $|M \triangle M'|$: the one-bead equations alone have the unique solution $w_N = \bar{w}_N = (-1)^N$; the two-bead equations alone have solution set $\{w_N = s^N, \bar{w}_N = s^{-N}\}$ for independent weights and $\{w_N = 1\} \cup \{w_N = (-1)^N\}$ for shared weights. Checked for $(e, f) \in \{(2, 3), (2, 4), (3, 4), (2, 5), (3, 5)\}$. This is Proposition 13 and Remark 14.
6. **The anchor identification, independently.** That $w_N = (-1)^N$ gives multiplication by p_e was checked without any ribbon, abacus or Murnaghan–Nakayama input: s_λ was expanded

in the monomial basis by counting semistandard tableaux (Kostka numbers), multiplied by $p_e = \sum_i x_i^e$ in $n = |\lambda| + e$ variables, and re-expanded in the Schur basis by inverting the (unitriangular) Kostka matrix. Agreement on 76/76 pairs with $e \leq 4$, $|\lambda| \leq 5$.

7. **The degenerate stratum.** Dropping the normalisation $w_0 = 1$ entirely and solving, the locus decomposes exactly as Theorem 7(a)–(c), with no further component: verified for $(e, f) \in \{(2, 3), (2, 4), (3, 4)\}$.

Negative controls

- (N1) **The solver manufactures nothing.** Run on $e = f = 3$, where the commutator is identically zero, the pipeline returns 0 equations and hence no constraint. (A pipeline that produced constraints here would be producing them from its own bookkeeping.)
- (N2) **The test is not degenerate in the new direction.** This is the control that matters, because the whole question is whether τ_2 is genuinely probed. For $(e, f) = (1, 3)$ the equation set is $\{\pm(w_1 + 1), \pm(w_1 + w_2)\}$; substituting the forced value $w_1 = -1$ leaves the residual equation $w_2 - 1$, which *does* contain w_2 . So w_2 is constrained by a relation that survives the determination of w_1 : the vanishing of the new freedom is measured, not assumed.
- (N3) **The deformation is real.** $R_3^w s_{(1)} = s_{(4)} + w_1 s_{(2,2)} + w_2 s_{(1,1,1,1)}$: the three weights sit on three distinct targets, so w_2 and w_1^2 are separated by the operator itself. A deformation invisible to the operator would make items 4 and 5 vacuous.
- (N4) **$[R_1, R_2]$ is excluded by construction.** It is nowhere used; it sees only w_1 and so cannot distinguish free weights from a single parameter.

7 What is and is not proved

Proved. Theorem 7 (all cases, all $1 \leq e < f$, independent weights, any integral domain), Corollaries 9 and 10, Proposition 13. Lemmas 4, 5, 11 are proved here; Lemma 12 is the free-weight form of [2, Thm. 5(2)] and is used only for Proposition 13 and for the sufficiency half of Theorem 7, where it is applied at the explicit weights $w_N = (-1)^N$.

Not proved / out of scope.

1. *Weights depending on more than the height.* A weight w_N could be replaced by a weight depending on the whole occupancy word of the interval $(b, b + e)$, not only on the number of beads in it. That is a strictly larger deformation (2^{e-1} parameters rather than e) and this note says nothing about it. The mechanism of Remark 15 suggests it will not help — the hinge/jumped dichotomy is a statement about one site — but the bookkeeping is different and I have not done it.
2. *q -deformed Clifford relations.* The deformation here is diagonal: the fermions stay undeformed and only the dressing moves. Deforming $\{\psi_a, \psi_b^*\}$ is a different question.
3. *Level $\ell > 1$.* Everything above is level 1.
4. *The adjoint relation.* I did not redo $[R_e^*, R_e] = [e]_{t^2}$ in the τ variables, as the brief permitted but did not require; it is not used anywhere here.

5. $e = f$. Trivially $[R_e^w, R_e^w] = 0$ for every w ; the theorem is about distinct sizes only, and this is not a gap but the hypothesis.

Effect on the earlier theorem. Nothing is retracted. The single-parameter forcing theorem — $R_e(t)$ and $R_f(t)$ do not commute for $e \neq f$, and $(1+t)$ divides every matrix element [1, Cor. 11], sharply [2, Cor. 7] — is the restriction of Theorem 7 to the curve $w_N = t^N$, and remains correct: there case (c) with $w_0 = 1$ reads $t^N = (-1)^N$, i.e. $t = -1$. What changes is its standing: the hypothesis “one parameter t ” was not load-bearing, and the conclusion “ $t = -1$ ” was a shadow of the sharper statement “the height weight must alternate”.

Remark 16. $R_e^{((-1)^\bullet)} = M_{p_e}$ is the Murnaghan–Nakayama rule $p_e s_\lambda = \sum_\mu (-1)^{\text{ht}(\mu/\lambda)} s_\mu$, verified independently in §6(6). The proof of Theorem 7 does not use it: sufficiency is checked directly on matrix elements, so the theorem is self-contained and the identification is a reading of the answer, not a step in it.

References

- [1] Clio, *A fermionic normal form for the ribbon operator $R_e(t)$, and its identification with the Kashiwara–Miwa–Stern boson*, 6 September 2026, `proofs/2026-09-06-c2-Q91-fermionic-normal-form.tex`.
- [2] Clio, *The cross-rank ribbon commutator*, 7 September 2026, `proofs/2026-09-07-Q92-cross-rank-commutator.tex`.
- [3] A. Lascoux, B. Leclerc, J.-Y. Thibon, *Green polynomials and Hall–Littlewood functions at roots of unity*, Lett. Math. Phys. **35** (1995) 359–374. (The move of replacing a parameter raised to a statistic by a parameter indexed by it.)